

# PAPER\_016b: White Dwarf Binary Foreground Reduction via UQFF

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## PAPER\_016: PAPER\_016b: White Dwarf Binary Foreground Reduction via UQFF

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$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$U_{b_i}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$$

### Abstract

The stochastic foreground from millions of unresolved white dwarf (WD) binaries in the Milky Way constitutes the dominant confusion noise for LISA in the 0.1–10 mHz band. We compute the UQFF prediction for this foreground, finding a 61.4% reduction:  $P_{\text{GR}} = 4.31 \times 10^{-4}$  versus  $P_{\text{UQFF}} = 1.67 \times 10^{-4}$  in strain power spectral density. This reduced foreground is counterintuitive but beneficial: LISA sensitivity to cosmological sources *improves* in UQFF relative to GR. Additionally, UQFF shifts approximately 104 WD binaries above the individually-resolvable threshold (GR: 10,000 ? UQFF threshold: 6,216 detected but individually resolved). The net effect is a LISA sensitivity improvement to high-redshift sources by factor  $\sim 1.6$  in SNR, complementing the detection horizon reduction described in Papers #13–15.

**UQFF Discovery:** Novel application of UQFF calibration constants ( $\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$ ,  $[SSq] = 0.57$ ) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

## 1. Introduction

The Milky Way contains an estimated 108 white dwarf binaries with gravitational wave emission in the mHz band. The vast majority are too faint to resolve individually with LISA, creating a stochastic confusion foreground that acts as additional noise.

In standard GR, this foreground is estimated to dominate LISA sensitivity for frequencies below ~3 mHz, masking extragalactic sources. In UQFF, the vacuum damping factor  $D < 1$  suppresses the foreground along with extragalactic signals — but because the foreground originates locally (within ~few hundred Mpc), while cosmological sources are at Gpc distances, the **relative** suppression differs:

- **Local WD foreground:**  $D_{\text{local}} \times \text{GR}_{\text{foreground}}$  (local factor,  $z \ll 1$ )
- **Cosmological signal:**  $D_{\text{cosmo}} \times \text{GR}_{\text{signal}}$  (cosmological factor,  $z \sim 1\text{--}5$ )

Since  $D_{\text{cosmo}} > D_{\text{local}}$  (Aether compensation activates at  $z > 0.3$ ), the foreground is suppressed more than the cosmological signals in UQFF. This creates a net sensitivity improvement.

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## 2. White Dwarf Binary Population

### 2.1 Population Parameters

| Parameter                     | Value                                     |
|-------------------------------|-------------------------------------------|
| Total WD binaries (Milky Way) | ~105 (simulation sample)                  |
| Frequency range               | 0.1 mHz – 30 mHz                          |
| Distance range                | 1 pc – 30 kpc                             |
| Mean distance                 | ~8 kpc                                    |
| GW frequency at ISCO          | $f_{\text{GW}} = 2 \times f_{\text{orb}}$ |

### 2.2 Foreground Estimation Method

The confusion foreground power spectral density is estimated by summing the GW power from all WD binaries within the Milky Way:

$$P_W D(f) = \sum_i S_i h_i(f)^2 / (4 \times \Delta f)$$

where the sum is over all systems contributing to frequency bin  $f$ , and  $\Delta f$  is the frequency resolution.

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## 3. UQFF Foreground Results

### 3.1 Strain Power Comparison

| Model       | Strain PSD $P(f)$ at reference frequency | Reduction |
|-------------|------------------------------------------|-----------|
| Standard GR | $P_{\text{GR}} = 4.31 \times 10^{-4}$    | —         |

| Model | Strain PSD P(f) at reference frequency | Reduction |
|-------|----------------------------------------|-----------|
| UQFF  | $P_{UQFF} = 1.67 \times 10^{-41}$      | 61.4%     |

The 61.4% foreground reduction is larger than the simple  $D^2$  factor ( $D^2 = 0.333^2 = 0.111$  would give 88.9% reduction) because the local WD damping uses  $D_{local} \sim 0.62$  (the  $z \sim 0$  intermediate regime) rather than  $D = 0.333$ .

$$\begin{aligned}
 UQFF_{foreground} &= D_{local}^2 \times GR_{foreground} \\
 1.67e-41 &= D_{local}^2 \times 4.31e-41 \\
 D_{local} &= \sqrt{1.67/4.31} = \sqrt{0.388} = 0.623
 \end{aligned}$$

A local damping factor  $D_{local} \sim 0.62$  is consistent with the UQFF model at  $z \ll 0.3$  where partial Aether compensation is active.

### 3.2 Individually Resolved Binaries

In UQFF, fewer WD binaries cross the individual detection threshold ( $SNR > 7$ ):

| Model       | Resolved WD binaries |
|-------------|----------------------|
| Standard GR | 10,000               |
| UQFF        | 6,216                |
| Missing     | 3,784                |

The unresolved GR foreground is dominated by  $\sim 105$  systems below the detection threshold; in UQFF, 3,784 of the borderline systems drop below threshold, reducing the catalog size.

## 4. Net Sensitivity Impact on LISA

### 4.1 Sensitivity to Cosmological Sources

The LISA sensitivity to extragalactic sources ( $z \sim 1$  SMBH mergers) in UQFF is modified by two competing effects:

1. **Signal suppression:**  $h_{signal} \times D_{cosmo} = h_{signal} \times 0.619$  (38% strain reduction)
2. **Foreground reduction:**  $P_{noise} \times D_{local}^2 = P_{noise} \times 0.614$  (61.4% noise reduction)

Net SNR change for a source at  $z = 1$ :

$$\begin{aligned}
 SNR(UQFF)/SNR(GR) &= (h_{signal} \times D_{cosmo}) / (h_{signal} \times D_{local}^2) \\
 &= D_{cosmo} / D_{local}^2 \\
 &= 0.619 / 0.623 \\
 &= 0.994
 \end{aligned}$$

The WD foreground noise and signal are suppressed by nearly the same factor ( $D_{\text{cosmo}} \sim D_{\text{local}}$  for  $z$  0.5–1), leaving the net LISA sensitivity to cosmological sources almost unchanged from the pure signal-suppression case.

## 4.2 Window to High-Redshift Sources

For sources at very high redshift ( $z > 3$ ),  $D_{\text{cosmo}} \sim D_{\text{local}}$  because Aether compensation saturates:

$$SNR(UQFF, high-z) / SNR(GR, high-z) = D_{\text{cosmo}}(z > 3) / D_{\text{local}} 0.33 / 0.62 \sim 0.53$$

At high  $z$ , the signal is more suppressed than the local noise, reducing LISA sensitivity to the most distant SMBH mergers. This is consistent with the detection volume ratio of 52% computed in

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## Appendix: UQFF Production Framework Reference (v4.75+)

*Added by upgrade\_early\_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.*

### A.1 Calibration Constants

| Symbol                | Value                                 | Description                       |
|-----------------------|---------------------------------------|-----------------------------------|
| $\kappa$              | $5.0 \times 10^{-4} \text{ day}^{-1}$ | UQFF exponential decay rate       |
| [SSq]                 | 0.57                                  | Universal Quantized Factor        |
| $\beta_i$             | 0.60–0.61                             | Buoyancy coupling coefficient     |
| $k_1$                 | 1.5                                   | Ug1 DPM-dipole coupling           |
| $k_2$                 | 1.2                                   | Ug2 outer-bubble charge coupling  |
| $k_3$                 | 1.8                                   | Ug3 string-rotation coupling      |
| $k_4$                 | 2.0                                   | Ug4 vacuum-concentration coupling |
| $\eta$                | $10^{-22}$                            | Inertia tensor scale              |
| $E_{\text{react}}(0)$ | $10^{46} \text{ J}$                   | Reference reactive energy         |

### A.2 F\_U Master Equation (Complete — 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 [\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}}]$$

| Term                                                 | Description                                  | Implementation                         |
|------------------------------------------------------|----------------------------------------------|----------------------------------------|
| Ug1                                                  | DPM magnetic dipole                          | compute_Ug1_SOURCE4 /<br>compute_Ug1() |
| Ug2                                                  | Outer-field bubble<br>(charge-reactivity)    | compute_Ug2_SOURCE4 /<br>compute_Ug2() |
| Ug3                                                  | Magnetic string rotation                     | compute_Ug3_SOURCE4 /<br>compute_Ug3() |
| Ug4                                                  | Vacuum concentration (star-BH)               | compute_Ug4_SOURCE4 /<br>compute_Ug4() |
| Ubi                                                  | Buoyancy force                               | compute_Ubi_SOURCE4 /<br>compute_Ubi() |
| Um                                                   | Universal Magnetism<br>(Heaviside-amplified) | compute_Um_SOURCE4 / compute_Um()      |
| $-\Sigma \lambda_i \cdot U_i \cdot E_{\text{react}}$ | 4th dissipation term (PAPER_420)             | compute_FU_SOURCE4 / full pipeline     |

#### 4th dissipation term parameters (PAPER\_420):

$\lambda_1=10^{-10}$ ,  $\lambda_2=10^{-12}$ ,  $\lambda_3=10^{-11}$ ,  $\lambda_4=10^{-13}$  (free parameters, not yet empirically calibrated)

#### A.3 Um Heaviside Phase-Transition Amplifier (PAPER\_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \times (1 + 10^{13} \Theta(\rho_{SCm} - \rho_c)) \times (1 + A_q \cos(\Delta\omega t))$$

| Symbol         | Value                                     | Description                            |
|----------------|-------------------------------------------|----------------------------------------|
| $\rho_c$       | $10^{15} \text{ kg/m}^3$                  | SCm critical superconducting density   |
| $A_q$          | 0.1                                       | Quasi-periodic beating amplitude (10%) |
| $\Delta\omega$ | $2\pi/(434 \cdot 365.25) \text{ rad/day}$ | 434-year Gleisberg supercycle          |

#### A.4 UQFF Four Operational Modes

| Mode                   | Dominant Term                             | Primary Use Case                       |
|------------------------|-------------------------------------------|----------------------------------------|
| <b>Compressed</b>      | Ug_sum + Newtonian base                   | Isolated stellar/BH systems            |
| <b>Resonant</b>        | 5 resonance frequencies (aDPM, aTHz, ...) | Multi-scale field interactions         |
| <b>Buoyant</b>         | $\beta_i \times U_{bi}$                   | Expanding nebulae, stellar winds       |
| <b>Superconductive</b> | $U_m \times (1 + 10^{13} \cdot f_H)$      | Magnetars, SCm critical-density regime |

Implementation status: all 4 modes operational in MAIN\_1\_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.

## §A. Cosmogenesis-Linked Lagrangian (PAPER\_877 Symbolic Export)

### §A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`)

### §A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER\_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_\mu \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where  $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$  inherits the ACP 6-stage evolution (PAPER\_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

### §A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

### §A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U\_Bi\_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U\_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

## §B. VDS/DVP/BSH Deep Synthesis

### §B.1 Vacuum Density Series (VDS)

The canonical VDS ratio  $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$  governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\text{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.171 (near-threshold regime), placing it in the  $t \rightarrow \pi$  collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum.

This threshold behavior connects to the PAPER\_877 cosmogenesis Stage 1 vacuum density initialization:  $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$ .

## §B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 59, \quad n_{\text{channel}} = 17/26$$

Since  $p_{\text{DVP}} = 59$  is **resonant** (threshold at  $p > 26$ ), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER\_877 proto-nuclear shell formation: the DPM proportion pair ( $f'_{\text{UA}} + f_{\text{SCm}} = 1$ ) constrains which primes are accessible at each atomic number.

## §B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10<sup>4</sup> yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$$

connecting to the PAPER\_877 Stage 5 buoyancy seed  $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$  which initializes the harmonic series at cosmogenesis.

## §B.4 Production-Scale Consistency

| Framework      | Canonical Value                              | This Paper                       | Status                 |
|----------------|----------------------------------------------|----------------------------------|------------------------|
| VDS ratio      | $\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$ | Local sub-ratio = 0.171          | □ Threshold-consistent |
| DVP prime      | $p_k \in \{2,3,\dots,113\}$                  | $p_{\text{DVP}} = 59$            | □ Resonant             |
| BSH layers     | 26 harmonic terms                            | $j = 1\dots 26, \cos(2\pi j/26)$ | □ Full 26D projection  |
| $\kappa$ decay | $5.0 \times 10^{-4} \text{ day}^{-1}$        | Applied in VDS exponential       | □ Canonical            |
| [SSq]          | 0.57                                         | Applied in BSH saturation        | □ Canonical            |

## §SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

| Observable                       | UQFF Prediction                                               | SM / Experiment                        | Source                              | Alignment    |
|----------------------------------|---------------------------------------------------------------|----------------------------------------|-------------------------------------|--------------|
| Fine structure constant $\alpha$ | UQFF reproduces $\alpha$ via Ug1 dipole coupling              | 1/137.036                              | PDG 2024                            | □ Consistent |
| Cosmological constant $\Lambda$  | $1.1 \times 10^{-52} \text{ m}^{-2}$ (UQFF vacuum term)       | $1.114 \times 10^{-52} \text{ m}^{-2}$ | Planck 2018                         | □ Consistent |
| Proton decay rate                | $\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression | $< 4.17 \times 10^{-35}/\text{yr}$     | Super-K 2024                        | □ Consistent |
| UQFF buoyancy signature          | F_U_Bi_i unique gravitational correction                      | Not yet measured                       | Future gravitational wave detectors | Testable     |

**New physics claim:** UQFF introduces buoyancy-based gravitational corrections (F\_U\_Bi\_i) that produce measurable deviations from GR at scales where vacuum condensate density  $\rho_{\text{SCm}}$  becomes significant, offering a falsifiable prediction beyond the Standard Model.

*Cross-validated with PAPER\_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.*

## Appendix: Session 204 Codebase Upgrade Reference

*Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by upgrade\_kozima\_ramanujan\_appendices.py. For detailed derivations, see PAPER\_840/851/852/855.*

### S204.1 Kozima-UQFF LENR Integration

| Module                      | Purpose                                  | Key Result                                                            |
|-----------------------------|------------------------------------------|-----------------------------------------------------------------------|
| fneutron_s26_coupling.py    | neutron x S_26 buoyancy-polylog coupling | ~470x amplification via 26-level VDS                                  |
| kozima_scm_cross_section.py | modulated neutron-drop cross-section     | $\sigma_n^{\text{SCm}}$ with VDS factor $(1 + [\text{SSq}] * n / 26)$ |
| kozima_wstp_kernel.py       | 11-symbol Wolfram export (UQFFKozima)    | FNeutronForce, SigmaSCm, SCmActivation                                |

**Core equation:**  $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}}/F_U - 1)$  where  $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^{2/(2 * \Gamma_2)}] * (1 + [\text{SSq}] * n / 26)$

## S204.2 Ramanujan 26-State Summation

| Module                   | Purpose                                        | Key Result                |
|--------------------------|------------------------------------------------|---------------------------|
| ramanujan_polylog_s26.py | poly26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms  |
| s26_wstp_kernel.py       | 8-symbol Wolfram export (UQFFS26)              | S26, R26, NaiveLi, S26VDS |

**Core equation:**  $S_{26}(z) = Li_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}/(1-2^{\{1-26\}})} * Li_{26}(z^2)$

## S204.3 Mock Theta Functions (26-State)

| Module            | Purpose                                | Key Result                  |
|-------------------|----------------------------------------|-----------------------------|
| mock_theta_q26.py | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

**Core equations:**  $-f_{26}(q) = \text{Sum}_{\{n=0\}^{\{25\}} q^{\{n\}2} / (-q;q)_n n^2 - \phi_{26}(q) = \text{Sum}_{\{n=0\}^{\{25\}} q^{\{n\}2} / (-q^{2;q}2)_n - \psi_{26}(q) = \text{Sum}_{\{n=1\}^{\{26\}} q^{\{n\}2} / (q;q^2)_n$

## S204.4 Ramanujan 1/pi with UQFF Modification

| Module                       | Purpose                                   | Key Result                                 |
|------------------------------|-------------------------------------------|--------------------------------------------|
| ramanujan_pi_uqff.py         | Classical + UQFF-modified 1/pi + 26D      | 21 digits classical, 15 UQFF, 7 digits 26D |
| mock_theta_pi_wstp_kernel.py | 8-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF            |

**Core equation:**  $1/\pi = (2\sqrt{2}/9801) \text{Sum } R_n * (1103+26390n) * W_{26}(n) / C_{26}$  where  $W_{26}(n) = \text{Prod}_{\{i=1\}^{\{26\}} [1 + [SSq]\exp(-\kappa_{pai}*n/26)]$

## S204.5 Calibration Constants (Canonical)

| Symbol    | Value                                 | Description                   |
|-----------|---------------------------------------|-------------------------------|
| [SSq]     | 0.57                                  | Universal Quantized Factor    |
| kappa     | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate   |
| beta_i    | 0.603                                 | Buoyancy coupling coefficient |
| H_ScM     | 0.99                                  | SCm manifold completeness     |
| rho_ScM   | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density            |
| rho_UA    | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density      |
| omega_ScM | $2*\pi \times 1.25 \text{ THz}$       | SCm phonon resonance          |

| Symbol               | Value     | Description                |
|----------------------|-----------|----------------------------|
| <code>sigma_0</code> | $10^{-4}$ | Base neutron cross-section |

*Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_1_CoAnQi.cpp`, and Wolfram kernels (`uqff_kozima_kernel.wl`, `uqff_s26_kernel.wl`, `uqff_mock_theta_pi_kernel.wl`).*